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A Security Enhancement Scheme for Image Perceptual Hashing

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***Abstract*—Perceptual hashing as a new technique for the multimedia content security have been widely discussed. However, the robustness requirement of perceptual hashing brings big challenges to its security. In this paper, we propose a secure enhancement scheme for image perceptual hashing based on error correction coding. By employing error correcting code, images with similar perceptual contents can be mapped into the same hash sequence. Consequently, we no longer store original hash sequence in the database of an image authentication system but the derived redundancy-check bits and cipher of perceptual hash. Our scheme can effectively resist the adversary from finding the relationship between the input image and its hash sequence. Therefore, the chosen-plaintext attack can be avoided. Although our scheme may sacrifice some robustness of an image perceptual hashing algorithm, both theoretical analysis and experimental results illuminate that the robustness loss is still acceptable.**

Keywords-perceptual hashing; chosen-plaintext attack; error correcting code; multimedia content security

I. INTRODUCTION

Traditional cryptographic hash (such as MD5, SHA1) is introduced to map the data to a short bit string called hash value or a message digest and it is sensitive to even single-bit variation of input data. The concept of perceptual hashing (also called robust hashing) is derived from traditional cryptographic hash function, encoding the perceptual features of multimedia content into a short hash vector. However, different from traditional data hash frangibility, robust hashing functions are designed to reflect perceptually modifications regardless the data format transformations or other content-preserving manipulations.

A typical image authentication process includes three steps, as shown in Figure 1, extracting features based on perceptual content, coding these features to generate hash, and then matching the hash to validate the credibility of received image. Similar to traditional data hash, this process is desired to satisfy some security properties, such as one-way and collision-free properties [1], and the confusion and diffusion capabilities [2]. In [1], the authors pointed out that, if the method of feature extraction is publicly known, the generated hash vector is vulnerable. It is very likely that an adversary could use a careful local forgery attack to fool the content authentication system. Therefore, to prevent forgery attack, the researchers introduced a secret key in feature extraction to permute the hash vector such as some well-known algorithms proposed by Venkatesan[3], Fridrich [4] and Wishal Monga[5]. The hash values could not be easily forged or estimated when the adversary without any knowledge of the secret key. However, Baris Coskun in [2] has proved that the randomness based on keys only contributes to the confusion capability and does not have satisfactory diffusion. In another word, robust hash is suffered to collision.

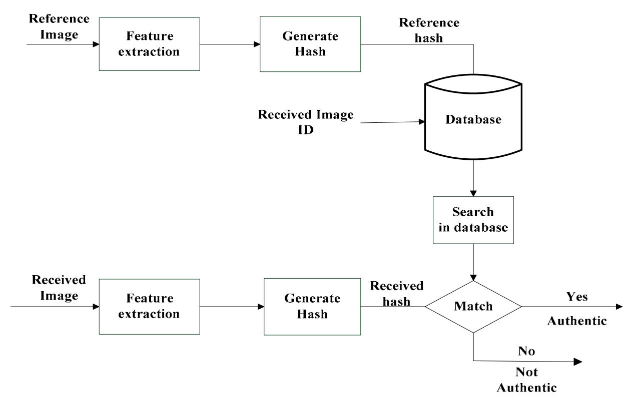

Figure 1. Perceptual hashing for Image Authentication

In fact, weak diffusion is due to robustness requirement of perceptual hashing. When the adversaries know the image and plaintext hash, they could use chosen-plaintext to obtain a great deal of information about the relation between the input and the hash value [2]. Thus, an effective way to resist this attack is hiding plaintext of perceptual hash. However, if we directly store encrypted hash sequence in the database, it is an issue for key management. In addition, the plain text of hash will still expose to adversary during decipher authentication phase. Based on these conflicts, in the paper, we propose a security hashing scheme. Our scheme introduces error correcting code to map hash sequences generated by the similar perceptual content images to the same hash value, and then execute the value hiding operation.

II. PROPOSED SCHEME

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A. Our Scheme

In figure 2, it shows the integral frame of the scheme. The generated hash that we store in the database can be described as follows:

1) Selecting a perceptual hashing function $PH(.)$ to extract hash sequences h_1 from the image I_1 in original images set.

$$h_1 = PH(I_1) \tag{1}$$

2) According to length of hash sequences h_1, we choose (n,k,t) linear block code and intercept the first k bits of h_1, denoting as $h_1^{'}$, to execute error correcting encoding $Ec(.)$, where k is close to the length of hash sequence, n is the length of error correcting code which can be divided into k bits generated sequence $h_1^{'}$ and redundancy check bits r_1, t is error-correcting number.

$$(h_1^{'} \mid r_1) = Ec(h_1) \tag{2}$$

3) Generating H_1 by hiding operations $En(.)$ .We could choose directly hash $h_1^{'}$ with traditional hash function (such as MD5, SHA1) as hiding operations.

$$H_1 = En(h_1^{'}) \tag{3}$$

4) Storing generated H_1 and redundancy check bits r_1 in the database.

For the authentication of a received image, our scheme also performs by several steps:

1) For the received image I'_1, search the database so as to confirm its original image through the ID information and then return corresponding redundancy check bits r_1.

2) Use the same robust hash function to generate perceptual hash value h_2, and then employ existed redundancy-check bits r_1 to correct $Dc(.)$ the first k bits of hash sequence h_2 in order to acquire corrected hash sequences $h_2^{'}$.

$$h_2^{'} = Dc(h_2 \mid r_1) \tag{4}$$

3) Similarly, hash $h_2^{'}$ in the same way to generate H_2.

$$H_2 = En(h_2^{'}) \tag{5}$$

Finally, by comparing these two sequences H_2 and H_1 the same or not, we can achieve the authentication result.

B. Security

In this sub-section, we discuss the security of our proposed scheme. Error correcting code has been used in traditional communication channel in order to reduce the transmission error rate. Within its fault tolerance, error correcting code can be used for accurate sequence transmission. On the other hand, because of robustness requirement, perceptual hash function needs to map hash sequences generated by the same perceptual content image set within a threshold. For example, hash functions reach the statistically irrelevance which corresponds to the normalized hamming distance (BER) equals to 0.5. Thus, if we use error correcting code to modify the changes of hash value in this image set, it means we could substitute this threshold to a fixed hash value.

Normally, in order to enhance security, most of image hashing algorithms use a secret key to randomize hash code in feature extraction stage. For instance, the Fridrich's method [4] projects image blocks onto pseudo-randomized low-pass images; and the well known Venkatesan's robust hash algorithm [3] extracts image features from randomly chosen regions of the image. However, as the author proposed in paper [2], when the adversary knows both the image and plaintext hash, they can obtain the information about secret key by chosen-plaintext attack, namely through observing changes in the hash value as the input is slightly changed they can evaluate the relation between the input image and each randomized hash bit, and then acquire the information about secret key.

Compared figure 2 with original verification process, the highlight change is the employment of error correcting code in our scheme. As this paper related, the significance of error correcting code is that it maps the images with the same perceptual content to a fixed point. In this way, we could no longer use hash matching but validate if the received image has the same hash result with original one. Additionally, the mapping employs perceptual hash to possess a lot of security properties as traditional hash which bring more benefits to hash hiding operations in applications. With the existence of error correcting code, the adversaries always see a single cipher rather than plain hash in database. Even if the error cannot be corrected, according to weak collision of traditional hash, the hash results in Equation (5) will be absolutely irrelevance with original image hash. Therefore, adversaries are unable to obtain the actual hash value and observe the slight changes in hash value by modifying image content, it will be difficult for them to carry out chosen-plaintext attack.

Our scheme enhances security by stopping the adversary from obtaining the hash plaintext, yet error correcting code makes the algorithms loss robustness. We use Y. Bian [6] 256-bit perceptual hash algorithms and then select its first 250 bits using (511,250,31) BCH linear block code to illuminate this effect. As shown in figure 2, when first k bits of generated from hash sequence h_2 cannot be corrected, the received image will be regarded as unauthentic. In fact, for (n,k,t) linear block code, t as the maximum hamming distance can be corrected and t/k corresponds to critical permission BER. In figure 3, hamming distance between "Lena" image is noised by variance 0.9 and original hashing is more than 31(corresponding critical permission BER is 0.124), which is to say, it will not be corrected to original hash value and pass validation in our scheme. But for the 256-bit generated perceptual hash sequence, the algorithm's

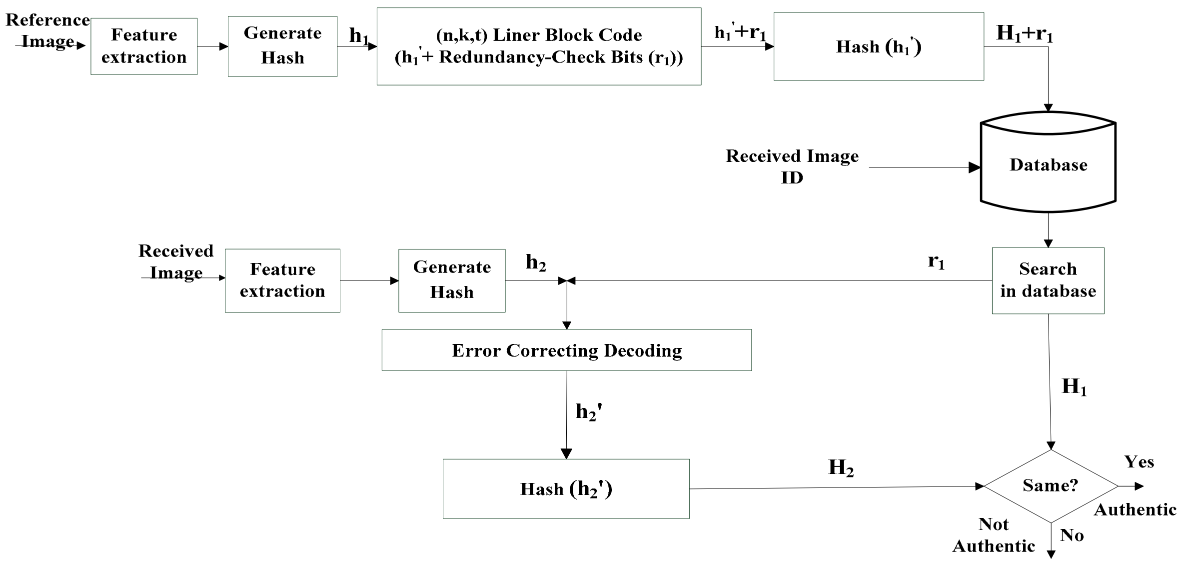

Figure 2. Our scheme to enhance the security of perceptual hashing

BER under variance 0.9 still less than 0.5 which means they do have the same perceptual content with original matching method.

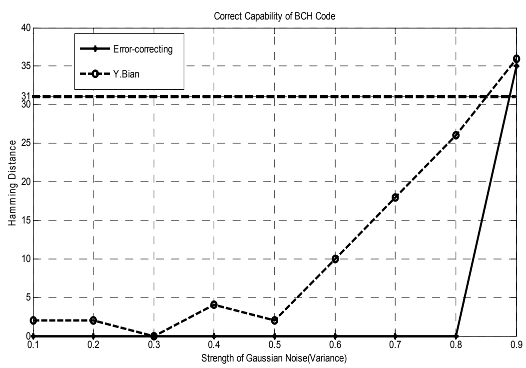

Figure 3. Correcting capability of BCH under Gaussian Noise (Variance)

III. APPLICATION AND EXPERIMENT

In this section, we will present a number of experimental results concerning if the losing robustness is acceptable, and then discuss the adaptability of our scheme. The whole experiment is performed on a picture database from [7] includes 44 images and many attack methods including Gaussian Noise, JPEG compression, blur, median filtering, and mosaic. We sequentially chose (511,250,31) BCH linear block code and Y. Bian [6] 256-bits perceptual hash algorithm which is based on the gray information in each divided block.

First of all, we use a group of inter-class test to gain the false acceptation rate (FAR) in order to check out the discriminability of our scheme. For the 703 samples inter-class test, the scheme has most zero percentage FAR. For intra-class robustness test, as the strength of distortion for each attack method is increased, the distorted image trends to be unrecognizable to the original gradually. With traditional matching method based on hamming distance, when two images with different perceptual content their BER is expected to be close to 0.5. Thus, when strength of distortion is increased to force the BER close to 0.5, we could achieve Critical Attack Strength (CAS). As the paper mentioned in section 2, fault tolerance number of (n,k,t) linear block code is t and t/k is the critical permission BER. When BER is increased exceeding t/k the image cannot pass validation in our scheme, thus we could also acquire Error Corrected Critical Attack Strength (ECCSA). In Table 1, we present Y.Bian algorithm CSA and ECCSA after error corrected. (For perceptual distance always smaller than 0.5, we record the BER under the worst effect of the attacks.)

TABLE 1. COMPARISON BETWEEN TESTED ALGORITHM AND CORRECTED STRENGTH THRESHOLD

Attack Method	Y.Bian (CSA)	Error-correcting (ECCSA)
Jpeg	0.0504 (Quality 1)	Quality 1
Gaussian Noise (mean)	0.1170 (Mean 0.95)	Mean 0.95
Median Filtering (times)	0.0115 (Times 50)	Times 50
Gaussian Noise (variance)	0.2197 (Variance 0.95)	**Variance 0.7**
Blur by Radius 5(times)	0.0818 (Times 40)	Times 40
Mosaic with Window Size	0.1487 (Window Size 60)	**Window Size 50**

As it shown in table 1, only CSA of Gaussian Noise (variance) and Mosaic are modified. The experiment illumines that if the robustness of algorithms for different attack methods meet with (n,k,t) linear block code fault tolerance requirement, our scheme does not affect their robustness. In figure 4, we present 512*512 baboon images under Gaussian Noise (variance) and Mosaic ECCSA. Through the observation of images, most of them have poor visual quality and the details are difficult to identify. Obviously, the robustness of algorithm is acceptable.

(a) Baboon image (b) Gaussian Noise (Variance 0.7) (c) Mosaic (Window Size 50)

Figure 4. Baboon images under part of ECCSA

However, not all of the algorithms are suitable for this scheme. We use another 256-bit image hash function that proposed by Fridrich's [4] to test adaptability of our scheme. The results show that the author's hash function has a bad discriminability since its FAR of inter-class is up to 0.5292. As shown in Figure 5, we compare the robustness of Fridrich's and Y.bian's perceptual hash algorithms under Gaussian Noise (by mean). The paper has already presented that (511,250,31) BCH linear block code its critical permission BER is 0.124, but the images noised by mean 0.05 would be regarded as unauthentic by the Fridrich for its BER larger than 0.124. We can say that, Fridrich is too strict for our scheme for the images noised by mean 0.05 still has good quality.

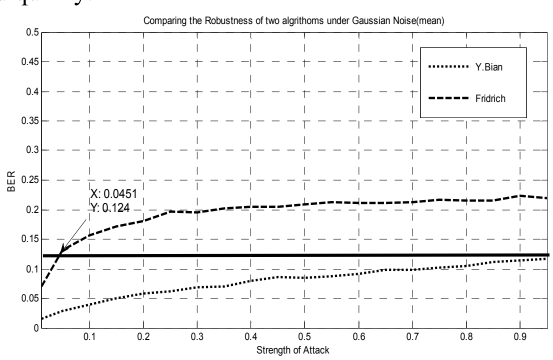

Figure 5. The two algorithms under Gaussian Noise (by mean)

The performance of the scheme can be determined by the robustness of perceptual hashing algorithms themselves. On the other hand, we could adjust the parameter k to reduce the losing robustness. For a fixed n in (n,k,t) linear block code, if the number of k that we use in hash sequences h_1 decreases, the corresponding critical permission BER will increase. For example, compare (511,250,31) and (511,241,36) of BCH linear block code, its critical permission BER moves from 0.124 to 0.149.

IV. CONCLUSION

According to published literature, all perceptual hashing based image authentication systems store the hash value of each image in the database. While, due to the weak diffusion capability of current image perceptual hashing functions, it may happen with high probability that the adversary can attain enough information about the relation between the input image and its hash value by the chosen-plaintext attack. In this paper, we proposed a secure enhancement scheme for image perceptual hashing, especially in the image authentication scenario. By using error correcting code, our scheme does not need to store the hash value any more. Since it is computation infeasibly for the attacker to derive the hash value from its redundancy-check bit, our scheme can effectively resist chosen-plaintext attack and improve the security of the perceptual hashing greatly. Through the theoretical analysis and a series of experiments, the feasibility of our scheme has been demonstrated. In our future work, we will concentrate on how to choose the optimal error correction code for a specific image perceptual hash function to minimize the robustness loss.

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